African Institute for Mathematical Sciences
RWANDA

# Introduction to Probability and Statistics

Lecture 8: Point Estimation

evans.gouno@univ-ubs.fr

## Introduction

- We observe $n$ r.v.'s $X_1, \ldots, X_n$ (not necessarily independent).

  Suppose that $X_i \sim f(x \mid \theta)$ where $\theta$ is an **unknown parameter**.

- The problem of estimating $\theta$ is a **problem of point estimation**.

### Definition

- *Any function of the r.v.'s of $X_1, \cdots, X_n$ is called a **statistic**.*
- *An **estimator** of $\theta$ is a statistic of $X_1, \cdots, X_n$:*

$$g_n(X_1, \cdots, X_n).$$

## Estimator

- Since $X_1, \cdots, X_n$ are r.v.'s, so is $g_n(X_1, \cdots, X_n)$.
- In general $\theta \in \Theta \subseteq \mathbb{R}^q$, that is $\theta = (\theta_1, \ldots, \theta_q)$.

  Examples: normal distribution: $\theta = (\mu, \sigma^2)$, Gamma distribution: $(\alpha, \beta)$.
- Vocabulary:
  - Estimator: $g_n(X_1, \cdots, X_n)$.
  - Estimate: $g_n(x_1, \ldots, x_n)$ (value).
- How could we build estimators?

  ➥ Methods of estimation?

## Empirical moments v. Theoritical moments

Let us consider $n$ r.v.'s $X_1, \cdots, X_n$.

- Empirical moment of order $k$:

$$m_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k \quad (statistic).$$

- Theoritical moments of order $k$

$$E(X^k;\theta) = \begin{cases} \displaystyle\sum_{x\in V} x^k P(X = x \mid \theta) & \textit{if X is discrete.} \\ \displaystyle\int_V x^k f_X(x \mid \theta)dx & \textit{if X is continuous.} \end{cases}$$

## Method of Moments Estimators (MME)

The method of moments estimators (MME) of $\theta_1, \dots \theta_q$ are the solution of the equations:

$$m_k = E(X^k; \theta_1, \dots \theta_q); \;\; k = 1 \dots, q.$$

Thus, if the parameter to be estimated is a $q$-dimensional vector, the method of moments propose to equate the first $q$ empirical moments to the first $q$ theoritical moments and to consider the solution with respect to $\theta_1, \dots, \theta_q$ as estimator of $\theta_1, \dots, \theta_q$.

**Exercise:** Compute the MME of the parameter of the following model given that a $n$-sample $X_1, \cdots, X_n$ is observed:

- $\mathscr{B}(N, p)$,
- $\mathscr{N}(\mu, \sigma^2)$,
- $\mathscr{G}(\alpha, \beta)$.

## Method of Moments

| **Distribution** | **Parameter** | **MME** |
|---|---|---|
| Bernoulli | $p$ | $\hat{p} = \overline{X}_n$ |
| Binomiale | $(N, p)$ | $\hat{p} = \overline{X}_n / N$ |
| Poisson | $\lambda$ | $\hat{\lambda} = \overline{X}_n$ |
| Exponential | $\theta$ | $\hat{\theta} = \overline{X}_n$ |
| Gamma | $(\alpha, \beta)$ | $\hat{\alpha} = \overline{X}_n^2 / s_X^2 \quad \hat{\beta} = \overline{X}_n / s_X^2$ |
| Normal | $(\mu, \sigma^2)$ | $\hat{\mu} = \overline{X}_n \qquad \hat{\sigma}^2 = s_X^2$ |

## Maximum Likelihood Method

- We observe $n$ independent r.v.'s $X_1, \ldots, X_n$.

  Consider $f(x_1, \ldots, x_n \mid \theta)$ or $P(X_1 = x_1, \ldots, X_n = x_n \mid \theta)$, the distribution of $X_1, \ldots, X_n$, where $\theta$ is an **unknown parameter**.

- $f(x_1, \ldots, x_n \mid \theta)$ or $P(X_1 = x_1, \ldots, X_n = x_n \mid \theta)$ can be seen as the chance of occurrence of $x_1, \ldots, x_n$.

  ➥ This is a function depending on $\theta$ which is an unknown parameter.

- A strategy to provide an estimator is to find the value of $\theta$ that maximize this quantity.

  ➠ What has been observed is the more likely.

## Maximum Likelihood Estimator

### Definition

*Let $(X_1, \cdots, X_n)$ be $n$ independent r.v.'s with distribution $f(x_1, \ldots, x_n \mid \theta)$ (or $P(X_1 = x_1, \ldots, X_n = x_n \mid \theta)$), where $\theta$ is an unknown parameter.*

- *The **likelihood function** is:*

$$L(\theta) = \begin{cases} f(x_1, \ldots, x_n \mid \theta) & (\text{continuous}). \\ \\ P(X_1 = x_1, \ldots, X_n = x_n \mid \theta) & (\text{discrete}). \end{cases}$$

- *Any estimator $g_n(X_1, \cdots, X_n)$ such that*

$$L\left(g_n(X_1, \cdots, X_n)\right) = \max\left\{L(\theta); \theta \in \Theta\right\}$$

*is called a **maximum likelihood estimator (MLE)**.*

## Maximum Likelihood Method

In practise, the procedure to obtain the MLE is the following:

1. Compute the likelihood:

$$L(\theta) = f(x_1, \dots, x_n \mid \theta) \left[ = \prod_{i=1}^{n} f(x_i \mid \theta) \textit{ when independence} \right]$$

$$\textit{or } L(\theta) = P(X_1 = x_1, \dots, X_n = x_n \mid \theta), \;\; \theta \in \Theta \subseteq \mathbb{R}^q.$$

2. Calculate the log-likelihood:

$$\log L(\theta).$$

3. Calculate the log-likelihood equations:

$$\frac{\partial}{\partial \theta_i} \log L(\theta) = 0, \;\; i = 1, \dots, q.$$

4. Solve these equations :

$$\hat{\theta}_n = \underset{\theta \in \Theta}{argmax} \log L(\theta).$$

## Maximum Likelihood Method

**Example : Binomial**
Let consider $n$ independent r.v.'s with a binomial distribution with parameter $(N, p)$.

1. Likelihood:

## Maximum Likelihood Method

**Example : Binomial**

Let consider $n$ independent r.v.'s with a binomial distribution with parameter $(N, p)$.

1. Likelihood:

$$L(p) = \prod_{i=1}^{n} \binom{N}{x_i} p^{\sum_{i=1}^{n} X_i} (1-p)^{nN - \sum_{i=1}^{n} X_i}.$$

2. Log-Likelihood:

## Maximum Likelihood Method

**Example : Binomial**

Let consider $n$ independent r.v.'s with a binomial distribution with parameter $(N, p)$.

1. Likelihood:

$$L(p) = \prod_{i=1}^{n} \binom{N}{x_i} p^{\sum_{i=1}^{n} X_i} (1-p)^{nN - \sum_{i=1}^{n} X_i}.$$

2. Log-Likelihood:

$$\log L(p) = \sum_{i=1}^{n} \log \binom{N}{X_i} + n\overline{X}_n \log p + n(N - \overline{X}_n) \log(1-p).$$

3. Likelihood equation:

# Maximum Likelihood Method

**Example : Binomial**

Let consider $n$ independent r.v.'s with a binomial distribution with parameter $(N, p)$.

1. Likelihood:

$$L(p) = \prod_{i=1}^{n} \binom{N}{x_i} p^{\sum_{i=1}^{n} X_i} (1-p)^{nN - \sum_{i=1}^{n} X_i}.$$

2. Log-Likelihood:

$$\log L(p) = \sum_{i=1}^{n} \log \binom{N}{X_i} + n\overline{X}_n \log p + n(N - \overline{X}_n) \log(1-p).$$

3. Likelihood equation:

$$\frac{\overline{X}_n}{p} - \frac{N - \overline{X}_n}{1-p} = 0.$$

4. Solution:

## Maximum Likelihood Method

**Example : Binomial**

Let consider $n$ independent r.v.'s with a binomial distribution with parameter $(N, p)$.

1. Likelihood:

$$L(p) = \prod_{i=1}^{n} \binom{N}{x_i} p^{\sum_{i=1}^{n} X_i} (1-p)^{nN - \sum_{i=1}^{n} X_i}.$$

2. Log-Likelihood:

$$\log L(p) = \sum_{i=1}^{n} \log \binom{N}{X_i} + n\overline{X}_n \log p + n(N - \overline{X}_n) \log(1-p).$$

3. Likelihood equation:

$$\frac{\overline{X}_n}{p} - \frac{N - \overline{X}_n}{1-p} = 0.$$

4. Solution:

$$\hat{p} = \overline{X}_n / N.$$

## Maximum Likelihood Method

**Example : Binomial**

Let consider $n$ independent r.v.'s with a binomial distribution with parameter $(N, p)$.

1. Likelihood:

$$L(p) = \prod_{i=1}^{n} \binom{N}{x_i} p^{\sum_{i=1}^{n} X_i} (1-p)^{nN - \sum_{i=1}^{n} X_i}.$$

2. Log-Likelihood:

$$\log L(p) = \sum_{i=1}^{n} \log \binom{N}{X_i} + n\overline{X}_n \log p + n(N - \overline{X}_n) \log(1-p).$$

3. Likelihood equation:

$$\frac{\overline{X}_n}{p} - \frac{N - \overline{X}_n}{1-p} = 0.$$

4. Solution:

$$\hat{p} = \overline{X}_n / N.$$

**Remark:** The MLE is the same as the MME.

# Maximum Likelihood Method

**Example : Exponential**
Let consider *n* independent r.v.'s with an exponential distribution with parameter $\theta$.

## Maximum Likelihood Method

**Example : Exponential**

Let consider $n$ independent r.v.'s with an exponential distribution with parameter $\theta$.

1. Likelihood

$$L(\theta) = \left(\frac{1}{\theta}\right)^n \exp\left\{-\frac{1}{\theta}\sum_{i=1}^{n} X_i\right\}.$$

2. Log-Likelihood

$$\log L(\theta) = -n\log\theta - \frac{1}{\theta}\sum_{i=1}^{n} X_i.$$

3. Likelihood equation:

$$-n + \frac{1}{\theta}\sum_{i=1}^{n} X_i = 0.$$

4. Solution:

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}_n.$$

**Remark:** The MLE is the same as the MME.

## Maximum Likelihood Method

**Exemple : Normal distribution**

## Maximum Likelihood Method

**Exemple : Normal distribution**

Let consider $n$ independent r.v.'s with a normal distribution with parameters $(\mu, \sigma^2)$.

1. Likelihood: $L(\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 \right\}$.

2. Log-Likelihood: $\log L(\mu, \sigma^2) = -n \log(\sqrt{2\pi}) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2$

3. Likelihood equations:

$$\begin{cases} \frac{\partial}{\partial \mu} \log L(\mu, \sigma^2) = 0 \\ \frac{\partial}{\partial \sigma^2} \log L(\mu, \sigma^2) = 0 \end{cases} \Leftrightarrow \begin{cases} \sum_{i=1}^{n} x_i - n\mu = 0 \\ -n + \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 = 0 \end{cases}$$

4. Solution: $\hat{\mu}_n = \overline{x}_n$ and $\hat{\sigma}^2 = s_x^2$.